
TRANSIENT AMPLIFICATION THROUGH NONLINEAR INHIBITION

A PREPRINT

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ABSTRACT

TBF

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1 Introduction

outside cortex, simple power law structure []

biophysics of single neuron [9] or disconnected neurons field models no space component [10, 3] traditional field model: Wilson and Cowan [12]

wilson cowan has been used for DBS [7]

Other modeling approach involve complex biophysical description (Einevoll’s models)

Microstim has been used to study attention (Moore Armstrong 2003), learning [?], perception [6] create brain computer interface (Roelfsema 2021)

Models non-Dale, originally one Gaussian [2] then mexican-hat [1]. Note that in [?] the long range inhibition is thought to be feedforward inhibition, not feedback inhibition. Different effective connectivity patterns are reviewed in [5].

For Dalean there is Wilson-Cowan [13, 11]. Wilson-Cowan (1973) which assumes I to E is wider than E to E. Somers et al. (1975) assumes only FF E (Center) and FF-I (surround) with surround connectivity wider than center. It’s not recurrent, the connectivity are for inputs from thal. Some recent work has compared with in vivo features [?], but here it is a gaussian E-E and uniform other connections.

Figure 1: Model tour

2 Results

3 Methods

Mathematical model description We consider a field model of DES with a small number of unknown parameters. The model aims at modeling the membrane potential of point-neurons covering the whole space around the electrode. In one dimension (x) and in a tissue with isotropic connectivity between excitatory and inhibitory factions ($w_{e \rightarrow e}(x)$, $w_{e \rightarrow i}(x)$, $w_{i \rightarrow i}(x)$ and $w_{i \rightarrow e}(x)$ capture the connectivity from any point of origin). The combination of the direct effect of electrical current (with intensity I) and the post-synaptic potentials from activated neurons near the electrode is modeled as

$$\frac{dv_e}{dt}(x, t) = -v_e + \delta(t)v_{ext}(x) + w_{ee} * \nu_e - w_{ei} * \nu_i, \quad (1)$$

$$\frac{dv_i}{dt}(x, t) = -v_i + w_{ei} * \nu_e - w_{ii} * \nu_i. \quad (2)$$

where $\nu_e = \Theta(v_e - v_T)$ is the firing rate, taken to be a nonlinear read out of the potential. Similalry, $\nu_i = \Theta(v_i - v_T)$ forms an equivalent relationship for the inhibitory population.

the first term of each equation encapsulate direct ephaptic effects of the stimulation while the second and third terms capture the effect of excitation and inhibition. In the following, we describe the ephaptic model and the connectivity in turn.

The resistance R lumps the effect of electric field angle, dielectric constant and the constant relating the external field to the membrane potential. $F(x)$ is a unitless decaying function of distance to the electrode, and is expected to be a power law with exponent -2 for a point current dipole and exponent -1 for a point monopole [8]. More precise modelling of electrode types [14] maybe used to substitute a different function of distance. In Eq. 1-2, we have added the parameter α with units of distance to take into account the fact that there can be no activation at a zero distance to the electrode. This is thought to reflect the spatial scale of cell-bodies and axons.

The connectivity is modelled with Gaussian relationship [4] with amplitude and reported in XXXXXX.

The activation radii ρ_e and ρ_i are by iteration of the threshold equation $V_e(\rho_e) = V_T$ implicitly.

3.1 Numerical simulation

The model can be simulated efficiently by noting the forward-Euler formulation of the time-discretized potential $v_e^{(i)}$ reduces to

$$v_e^{(i)}(x) = v_e^{(i-1)}(x) + \Delta t \left[-v_e^{(i-1)} + \operatorname{erf} \left(\frac{x + \rho_e^{(i-1)}}{\sigma_{ee}} \right) - \operatorname{erf} \left(\frac{x - \rho_e^{(i-1)}}{\sigma_{ee}} \right) \right] \quad (3)$$

$$K_\alpha(x; \rho_e) = \operatorname{erf} \left(\frac{x + \rho_e^{(i-1)}}{\sigma_{ee}} \right) - \operatorname{erf} \left(\frac{x - \rho_e^{(i-1)}}{\sigma_{ee}} \right) \quad (4)$$

for the case Gaussian connectivity and Heaviside link functions.

3.2 Choosing parameter values

Hand tuned to fit data, procedure: 0. theta and sigma fixed, so really only 3 free parameters. 1. R to fit slope along ephaptic line (after ephaptic rheobase but before pop rheobase), 2. alpha to fit pop rheobase, 3: tweak w_0 while compensating with α to get a fit.

4 Appendix

4.1 Derivation of numerical integration procedure

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